Speeding up the sampler: status and options

Notes for the supervisor meeting

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`my_extendedSampling` / `softmax`

2026–11–03

Abstract

Goal: make the gradient-informed sampler on the zero_polymer reference (influenza HA0, $L = 566$) fast enough to reach equilibrium at and below the folding crossover. One change has paid off. The *one-pass move* computes the exact energy and the gradient for all sites with one forward and one backward pass through ESM3. It costs 0.153 s per move instead of 0.365 s, a 2.38× speed-up with the same per-move statistics and an exact Metropolis–Hastings kernel. Three other ideas were measured and give little or nothing: running two chains on the GPU at once (−7%), mutating several sites per move ($\leq 1.16\times$, only at high T), and replica exchange (no gain per move at an affordable length). The limit now is acceptance, not cost: at $T \leq 15$ only 0.5–6% of moves are accepted. The most promising next step is an exact single-site move that evaluates all 19 alternatives in one batched forward (Metropolized Gibbs), worth up to $\sim 12\times$ at low T if batching is efficient. It is not built yet, and it removes the gradient from the proposal, which is the main decision for this meeting.

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1 Where the time goes

A move picks a site uniformly, proposes a new residue from a softmax over the gradient-predicted energy change (drift $-\Delta t^2 \nabla U/2M$, noise $\Delta t\sqrt{T/M}$; $\Delta t/\sqrt{T}$ is fixed across T, so the informedness is the same at every temperature), and accepts with the Metropolis–Hastings ratio. The energy is the attention-map distance to the reference,

$$U(s) = \sum_{i \leq j} \left[\ln A_{ij}(s) - \ln A_{ij}(s_{\text{ref}})\right]^2,$$

where $A(s)$ is the head-averaged attention map of ESM3's last transformer layer for sequence s alone (symmetrised, BOS/EOS removed; bf16). So $U(s_{\text{ref}}) = 0$ and $U \geq 0$; the structure cross-entropy and entropy terms are off ($\lambda_{\text{am}} = 1$, $\lambda_{\text{sce}} = \lambda_S = 0$). Every cost is GPU time on the RTX 4500 Ada (24 GB, "astaroth"), and the GPU is fully busy during a move.

move	network passes per move	s/move	10^5 moves
site_soft (before)	3 forward + 2 backward	0.365	10.1 h
one-pass (now)	1 forward + 1 backward	0.153	4.3 h

Why this matters: at $T = 100$ two reference-started and two random-started chains needed $\sim$140 000 one-pass moves ($\sim$6 h each) to agree on H_d ($\hat{R}_U = 1.48$, $\hat{R}_{H_d} = 1.13$). At $T = 32$ they still disagree after 140 000 ($\hat{R}_U = 5.25$). At $T = 12$ a single reference-started chain needed $\sim$400 000 moves ($\sim$17 h) to reach a quasi-plateau. With single-site moves at acceptance 0.08, touching each of the 566 sites about $\ln 566$ times already takes $\sim$45 000 moves.

2 What has been done

2.1 One-pass move: 2.38× per move (done, in production)

Two changes (details in `one_pass_move.tex`):

1. The gradient is taken with respect to the one-hot probability matrix at the exact sequence, so the forward pass that gives the exact U also gives the gradient.

2. The gradient is computed for all sites at once (same cost as for one site). The gradient at the candidate becomes the gradient of the next move if the candidate is accepted, so each move needs one forward + one backward.

The gradient changes scale but not its ranking of residues, so Δt is rescaled by a computable factor $c = p_{\text{off}}/T_{\text{sftm}} = 4.535 \times 10^{-4}$. The kernel is exact for any gradient, including the non-bit-reproducible bf16 backward.

test	what	result
T0	gradient vs finite differences (fp64)	median rel. error 4–5 $\times 10^{-6}$
T1	U from the grad-enabled forward	equal to the exact U bit for bit (50/50)
T2	one-pass vs site_soft gradient (fp32)	cosine ≥ 0.999997, scale $1.001\,c$
T4	Boltzmann on a 2-site restricted problem	KL/TV 0.0154/0.0527 (site_soft 0.0169/0.0569), both $\propto$ MC noise; acc. 0.305 vs 0.306
T5	informedness vs Δt	p_{match} agrees site by site to $\sim$1%
T6	cost	0.153 vs 0.365 s/move; peak memory 6350 MB in both
T7a	transient from the reference, $T = 32, 100, 316$	acceptance agrees ($\lvert z\rvert \leq 1.4$); H_d, U within a few %

Toy-energy checks (exact enumeration, $L = 4, 5$, $N = 1$–3): detailed balance to 3×10^{-18}; a deliberately wrong variant was detected (TV 0.050 vs 0.008).

2.2 Two chains on the GPU at once: no gain

Two chains together run at 0.329 s/move each (one-pass) or 0.79 s/move (site_soft): a move every 0.164 s instead of 0.153 s, i.e. 7–8% slower than running them one after the other. The scripts now run one job at a time.

2.3 Several mutations per move (n_{sites} scan): at most 1.16×

From the $T = 32$ and $T = 100$ chains at 140 000 moves, 5000 moves per N:

T	N	acceptance a_N	Na_N	s/move	s per decorrelation
32	1	0.159	0.159	0.153	> 384
32	2	0.051	0.102	0.154	> 385
32	4	0.004	0.016	0.154	–
32	≥ 8	0.000	0	0.156–0.164	–
100	1	0.497	0.497	0.153	173
100	2	0.283	0.567	0.154	**149** (1.16×)
100	4	0.135	0.541	0.154	173
100	8	0.038	0.302	0.156	266
100	16	0.003	0.053	0.158	> 396

The cost of a move barely depends on N (0.153 s at $N = 1$, 0.164 s at $N = 32$), so the problem is acceptance: $a_N \approx a_1^N$. The reason is that each site has its own uphill cost (median $\Delta U \approx 222$ for one mutation near the reference, far above T), and an N-site move must pay all N costs in one accept/reject. Site–site coupling is minor (13% of the summed single-site effect for dispersed pairs, 27% within 30 residues). With $a_N = a_1^N$ the best N is $\approx -1/\ln a_1$, so $N = 1$ wins whenever $a_1 < 0.37$, i.e. at every T that matters. Adaptive N or "choose weakly coupled sites" cannot beat the best fixed N.

Answer to the original question: one mutation per step at and below the crossover; two at high T, for 16%.

2.4 Replica exchange (parallel tempering) pilot: no gain at this length

16 temperatures from 12 to 316, 2000 sweeps (2.45 s per sweep). Swap acceptance between neighbours was 0.18–0.47, so the ladder works, and walkers travelled across it. But each rung was no closer to equilibrium than a plain chain after the same number of moves: swaps only help once the hot rungs are equilibrated, which here takes ∼20 000 sweeps (14 h per ladder, at 16× the cost per sweep). For three temperatures, plain chains are cheaper. Code kept (`run_pt.py`) for a temperature scan later.

2.5 What the faster sampler has produced

- **Crossover location.** H_d (Hamming distance to the reference) falls smoothly with T: ∼380 at $T = 12$, > 330 at $T = 18$, ∼495 at $T = 32$, ∼528 at $T = 100$ (random sequence: 538). A one-coordinate free energy $F(h) = U(h) - T\ln[\binom{L}{h}19^h]$ gives $T^*(h) = (dU/dh)/\ln[19(L-h)/h] \approx 7$–15 at $h \sim 300$, consistent with a gradual crossover around $T \approx 8$–15 rather than a sharp transition.

- **Low T.** $T = 12$ reference chain: 580 000 moves, H_d plateau ∼380, $U \approx 6.9\times10^3$, acceptance 0.05–0.06; being extended to 10^6 (ends today ∼10:00). $T = 5$ queued next. Random starts never reach the reference basin below the crossover (stuck at $H_d \approx 530$), so $\hat{R}$ between reference and random starts cannot be met there.

3 What limits speed now

1. **Acceptance at low T.** 0.5–4% at $T = 8$–15 at the start, 5–6% at the $T = 12$ plateau. More than 94% of the GPU time at low T is spent on rejected moves.

2. **The gradient is a weak guide at $L = 566$.** Its predicted sign of ΔU is right for 13 of 20 single mutations, and its top residue is the true best at 2 of 10 sites. The proposal is informed, but not by much.

3. **One chain does not fill the GPU (estimate).** One forward is ∼60 ms for ∼1.6 TFLOP, about 15–20% of the card's bf16 peak. Not yet measured; this decides between the options below.

4 Options from here

Gains are wall-clock accepted changes per second relative to today's one-pass move. t_{19} = time of one batched forward over 19 sequences, not yet measured (between ∼0.25 s with good batching and ∼1.1 s with none).

option	idea	expected gain	status / effort
A. Metropolized Gibbs (batched exact single-site move)	at a uniform site, exact U of all 19 alternatives in one batched forward; propose j with prob. $\pi_j/(1-\pi_{\text{cur}})$, accept $\min[1, (1-\pi_{\text{cur}})/(1-\pi_j)]$	$T = 8$: $\sim 12\times$ (good batching) or $2.6\times$ (none); $T = 32$: $3.8\times$ / $0.8\times$; $T = 100$: $1.2\times$ / $0.3\times$	∼1 day + toy Boltzmann test; needs t_{19}
B. Parallel chains in one batch	B chains share each forward and backward	total throughput $B\,t_1/t_B$, not a faster single chain; helps 4-chain $\hat{R}$ runs and PT	∼1 day; $B \leq 3$ on 24 GB
C. CUDA graphs / `torch.compile`	same move, less overhead	unknown, plausibly 1.3–2×; stacks with A and B	0.5–2 days
D. Gradient-chosen site (Gibbs-with-gradients)	choose site and residue from all $L \times 19$ predicted ΔU (already cached by one-pass)	small while the gradient ranks near chance	0.5 day
E. More sites per move	adaptive N, weakly coupled sites	$\leq 1.16\times$, only at high T	not recommended
F. Parallel tempering	production ladder with fewer rungs	useful for a T scan, not for 2–3 temperatures	code ready

The benchmark that decides. `code/tests/bench_batch.py` is written but not yet run (∼10 minutes on astaroth). It measures forward and forward+backward time for batches of 1–32 sequences (gives t_{19} and the parallel-chain throughput), the fraction of a forward during which the GPU is running kernels, and, at the $T = 12$ state, the exact ΔU of all 19 mutants at 20 sites (gives the gradient's ranking quality and the predicted Metropolized-Gibbs gain for $T = 8$–100). It is on the branch `claude/project-thread-uu4fbg`. Decision rules:

- $t_{19} \lesssim 0.3$ s (about 5× one forward): build A first.
- batching barely helps but the GPU is mostly idle: C first.
- gradient–exact Spearman > 0.5: keep the gradient as the site selector inside A (a hybrid of A and D).

Keeping the gradient. Option A needs no gradient: the reverse move uses the same 20 exact energies. The gradient-informed move is the subject of the thesis, so there are three ways to frame A:

1. A as a baseline: the exact local move against which the gradient move's informedness is measured (how close does the one-pass proposal get to the ideal single-site proposal?).
2. A hybrid: the gradient picks *which site* to update (option D), A picks the residue exactly.
3. A as the production sampler for equilibrium data at low T, with the gradient move kept for the method study.

5 Beyond speed: the structural check

The plan was to check that low-T samples keep the native fold (Rg, pLDDT from ESM3 structure decoding) and that low-entropy sites match sites conserved in nature. Both checks are inconclusive for HA0, for reasons outside the sampler:

- **The decoder does not fold the reference.** ESM3-open decodes the reference to pLDDT 0.51/0.60/0.62 and Rg 45/54/70 Å at 8/32/128 steps (0.61 at 200, 0.59 at 256). The ectodomain alone (16–528) is no better (pLDDT 0.51–0.63). The decode is deterministic, so the spread over $T = 12$ samples (Rg 42–65 Å) is real but not a fold readout. HA is a trimer, and ESM3 predicts it as a monomer.
- **The H3N2 alignment is too narrow.** 457 of 566 sites are below 0.1 bits; the variable ones are mostly antigenic. Spearman between sampled and H3N2 site entropy: 0.07 at $T = 12$, ≈ 0 at $T = 32$, 100. Overlap of the 10% most conserved sites: 8 vs 5.7 expected ($p = 0.2$). A multi-subtype HA alignment is being built.

Options for the structure: a multimer predictor (AlphaFold2-multimer via ColabFold, Boltz-1, Chai-1) run on exported sequences as a trimer, or ESM3 with the three chains joined by a 25-glycine linker as a quick test. Caveat: reusing the reference MSA for sampled sequences at ∼33% identity would bias predictions toward the native fold; single-sequence or MSA-free modes avoid this but are weaker.

6 Questions for the supervisors

1. Is it acceptable to use an exact batched move (Metropolized Gibbs) with no gradient in the proposal, as a baseline, a hybrid, or the production sampler at low T? (Pending the benchmark numbers.)
2. The $H_d(T)$ curve is a smooth crossover. Which temperatures should the final study use (current: 5, 12, 32, 100)?
3. Below the crossover, random starts never reach the reference basin. Is "a long reference-started chain with a flat H_d and U" an acceptable equilibrium criterion there, in place of $\hat{R}$ between independent starts?
4. Is HA0 the right test case, given that it is a trimer that the structure decoder cannot fold? A monomeric reference (e.g. protein G, already used) would make the structural check meaningful.
5. Which structure predictor, and is the MSA bias acceptable?

Immediate next steps

- Run `bench_batch.py` (10 min) between the end of $T = 12$ and the start of $T = 5$.

- $T = 12$ to 10^6 moves (ends today $\sim$10:00); rerun structure and site entropy comparison on the last 500 000 moves with the multi-subtype HA alignment; then resume $T = 5$.
- Build option A (or C) according to the benchmark and the answer to question 1.